G.C.E.(O/L) ICT – 2024(2025)

Tentative Marking Scheme

By,

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PART - I

Que No.	Ans	Que No.	Ans	Que No.	Ans	Que No.	Ans
01.	4	11.	1	21.	4	31.	4
02.	3	12.	2	22.	2	32.	1
03.	4	13.	3	23.	4	33.	2
04.	4	14.	2	24.	3	34.	?
05.	4	15.	2	25.	3	35.	3
06.	3	16.	2	26.	2	36.	4
07.	2	17.	3	27.	3	37.	3
08.	2	18.	3	28.	2	38.	3
09.	1	19.	3	29.	2	39.	1
10.	4	20.	4	30.	4	40.	4

(40 × 1 = 40 Marks)

PART II

1. (i) 2 Marks Shortest distance, Farthest distance, Total Distance, Time taken to travel among cities, Nearest Cities, Farthest Cities.

(Any two answers) -----(2)

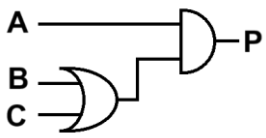
(ii) 2 Marks
(a) E -----(½)
(b) B -----(½)
(c) C -----(½)
(d) A -----(½)

(iii) 2 Marks
(a) 1001010_2 -----(1)
(b) 226_{16} -----(1)

(Should mention base value in answer)

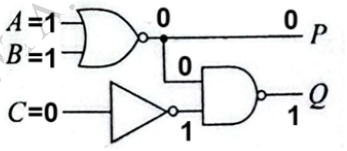
(iv) 2 Marks
(a)

P = A(B+C)



-----(1)

(b)



P = 0, Q = 1 -----(1)

(v) 2 Marks
(a)
A → 3 -----(¼)
B → 1 -----(¼)
C → 4 -----(¼)
D → 2 -----(¼)

(b)
A → 2 -----($^1/_5$)
B → 4 -----($^1/_5$)
C → 3 -----($^1/_5$)
D → 4 -----($^1/_5$)
E → 1 -----($^1/_5$)

(vi) 2 Marks
1 → E -----(½)
2 → C -----(½)
3 → B -----(½)
4 → D -----(½)

(vii) 2 Marks

START	-----(¼)
READ N1, N2	-----(¼)
IF N1 > N2 THEN	-----(¼)
PRINT "First is greater"	-----(¼)
ELSE	-----(¼)
PRINT "Second is greater or equal"	-----(¼)
END IF	-----(¼)
STOP	-----(¼)

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(viii) 2 Marks (a) In the context of software or system development, a **prototype** is an early, working model or sample of a product or system built to test a concept or process or to act as a thing to be replicated or learned from.

Specifically in the requirements identification phase, a prototype serves as a **tangible representation** of the system being envisioned. It's not the final product, but rather a simplified version designed to:

- **Visualize and demonstrate functionality:** It allows stakeholders (users, developers, clients) to see and interact with a representation of the system, even in its early stages.
- **Gather feedback and refine requirements:** By interacting with the prototype, users can provide valuable feedback, highlighting what works, what doesn't, and what's missing. This feedback helps refine and clarify the requirements.
- **Identify potential issues early:** Prototyping can reveal design flaws, usability problems, or misunderstandings about requirements before significant development effort is invested.
- **Improve communication:** Prototypes serve as a common ground for communication between stakeholders, fostering a shared understanding of the system's goals and features.

-----(1)

(b) A – Coding of the solution -----(⅓)
B – Testing and Debugging -----(⅓)
C – Deployment of the system -----(⅓)

(ix) 2 Marks (a) 10.0.0.1 ***or*** 172.16.0.1 ***or*** 8.8.8.8 ***or*** 192.168.15.8 (Any one suitable notation) -----(1)

(b) A – ✓ -----(⅓)
B – ✓ -----(⅓)
C – ✓ -----(⅓)

(x) 2 Marks (a) Regularly back up your files to an external storage device or a cloud-based service. -----(1)
(b) Take frequent breaks and stretch regularly / any suitable answers -----(1)

Part II, Q1 = 20 Marks

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2. (i) 1 Mark = C8*D8 (Not case sensitive) -----(1)

(ii) 2 Marks (a) = SUM(E8:E10) ***or*** = SUM(E8,E9,E10) ***or*** =E8+E9+E10
(Not case sensitive, any of one answer) -----(1)

(b) = SUM(E13:E18) ***or*** = SUM(E13,E14,E15,E16,E17,E18) ***or*** = E13+E14+E15+E16+E17+E18
(Not case sensitive, any of one answer) -----(1)

(iii) 1 Mark =E11-E19 ***or*** =SUM(E11,-E19) (Not case sensitive, any of one answer) -----(1)

(iv) 1 Mark =B4 -----(1)

(v) 2 Marks (a) → 1-Date -----(1) (b) → 2-Decimal Places & 5-Thousand Separator -----(1)

(vi) 2 Marks (a) Column Chart -----(1) (b) Pie Chart -----(1)

(vii) 1 Marks = C17^C9/2-(180-C10*3)
= 10^2/2-(180-50*3)
= 100/2-(180-150) = 50-30
=**20** -----(1)

Part II, Q2 = 10 Marks

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3. (i) *2 Marks*
CUSTOMER → (4005, Tom) -----(1)
RENTAL → (4005, V004, 20/02/2025) -----(1)

(ii) *2 Marks* VEHICLE, CUSTOMER and RENTAL ***or*** All three tables -----(⅔, ⅔, ⅔)

(iii) *2 Marks* VEHICLE table -----(2)

(iv) *2 Marks* RENTAL table -----(2)

(v) *2 Marks* (a) New table name → TECHNICIAN
Field Names on TECHNICIAN table → T_ id, T_Name, Phone_No -----(1)

or

TECHNICIAN

T_id	T_Name	Phone_No

(b) Need to add an additional field named "T_ID" to the VEHICLE table. -----(1)

or

VECHICLE

V_id	V_model	V_type	T_id

Part II, Q3 = 10 Marks

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4. (i) *4 Marks*
A → Yes -----(1)
B → No -----(1)
C → Sum = Sum + Number -----(1)
D → Is Number > 20 ***or*** Is Number >= 21 ***or*** Is Number = 21 -----(1)

(ii) *3 Marks*
(a) 1 2 3 4 5 ***or*** 1 -----(1)
2
3
4
5

(b) 1 4 7 10 13 16 19 ***or*** 1 -----(1)
4
7
10
13
16
19

(c) Loop never stops ***or*** Displays 1 2 3 4 …….∞ ***or*** Becomes an Endless loop -----(1)

(iii) *3 Marks*
A → 50 -----(½)
B → 0 -----(½)
C → Total -----(½)
D → Coin_value -----(½)
E → Price ***or*** 50 -----(½)
F → Price ***or*** 50 -----(½)

Part II, Q4 = 10 Marks

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5. (i) *10 Marks*

A → 4	-----(1)	F → 7	-----(1)
B → 2	-----(1)	G → 1	-----(1)
C → 5	-----(1)	H → 9	-----(1)
D → 3	-----(1)	I → 10	-----(1)
E → 8	-----(1)	J → 6	-----(1)

(ii) *4 Marks*

Student Name	Subjects
Alice	Math
	Science
Bob	English

-----(4)

(iii) *16 Marks*

① → <Head>	(1)	⑨ → <dl>	(1)
② → </Head>	(1)	⑩ → </dl>	(1)
③ → <Body>	(1)	⑪ → <dt>	(1)
④ → </Body>	(1)	⑫ → </dt>	(1)
⑤ → <h1>	(1)	⑬ → <dd>	(1)
⑥ → </h1>	(1)	⑭ → </dd>	(1)
⑦ →	(1)	⑮ → a	(1)
⑧ →	(1)	⑯ → 	(1)

Part II, Q5 = 30/3 = 10 Marks

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6. (i) *5 Marks*

(a) One good way to inform patients about available time slots is to use a **calendar-based interface with visual cues**.

- **Calendar View:** The system could display a calendar where patients can select a date.
- **Time Slot Availability:** For each date, the system could show the available time slots using different colors or icons. For example:
  - Green: Available
  - Yellow: Limited Availability
  - Red: Fully Booked
- **Interactive Selection:** Patients can click on a date and then select a specific available time slot from a list or a grid view.

This approach provides a clear and intuitive way for patients to see and select available time slots.

-----(1)

(b) Based on the notes (no doctor details, no login required), here are four main inputs:

1. **Patient's Name (or Identifier):** To identify the patient.
2. **Date of Appointment:** The specific date the patient wishes to visit the clinic.
3. **Time Slot:** The chosen time slot on that date.
4. **Reason for Appointment (or Department/Service Required):** To help the clinic prepare for the patient's visit and direct them to the appropriate area.

-----(1)

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(c) A suitable method would be to send an **SMS (text message) or an email confirmation** to the patient.

- **SMS Confirmation:** * **Pros:** Direct, immediate, high open rate, doesn't require internet access (beyond mobile network).
  - **Example:** "Your appointment at [Clinic Name] is confirmed for [Date] at [Time]. Please arrive 15 minutes early. [Clinic Phone Number]"
- **Email Confirmation:**
  - **Pros:** Can include more detailed information, such as directions, pre-appointment instructions, and attachments.
  - **Example:**
    - Subject: Appointment Confirmation - [Clinic Name]
    - Body: Dear [Patient Name], Your appointment at [Clinic Name] has been confirmed for [Date] at [Time]. Please find attached a map to our location and pre-appointment instructions. [Clinic Phone Number]

-----(1)

(d) The **Waterfall model** would be the most suitable in this scenario.

- **Waterfall Model:** This model follows a sequential, linear approach. Each phase (requirements, design, implementation, testing, deployment) must be completed before moving to the next.
  - **Why it's suitable:**
    - **Clear Requirements:** The problem states the requirements are clear and unlikely to change. Waterfall thrives when requirements are well-defined upfront.
    - **Small System:** Waterfall is often suitable for smaller projects where the complexity is lower.
    - **Low Risk:** The low likelihood of requirement changes reduces the risk of rework, which is a significant drawback of Waterfall in projects with evolving needs.
- **Iterative Incremental Model:** This model involves developing the system in increments, with each increment building upon the previous one. It's more suited for projects with evolving requirements or when early feedback is crucial.

-----(1)

(e) **Direct Deployment** is more appropriate in this scenario.

- **Direct Deployment:** The old system is completely replaced by the new system at a specific point in time.
  - **Justification:**
    - **Well-Tested System:** The problem states the system is "developed and tested well," reducing the risk of major issues during deployment.
    - **Simplicity:** Direct deployment is simpler and faster, requiring less overhead than parallel deployment.
    - **Cost-Effective:** It avoids the costs of running two systems simultaneously.
    - **Immediate Benefit:** Patients can start using the new system immediately, solving their appointment scheduling problems.
- **Parallel Deployment:** The old and new systems run concurrently for a period of time, allowing for comparison and a gradual transition. This is typically used when the new system is complex or has a high risk of failure.

-----(1)

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(ii) *2 Marks* Here are two changes to improve the situation described in the text:

1. **Create Separate User Accounts:** Each person using the computer should have their own user account with unique login credentials. This would:
    - **Isolate Files:** Prevent accidental modification or deletion of files by other users.
    - **Personalize Settings:** Allow each person to customize their desktop, applications, and settings to their preferences.
    - **Improve Organization:** Ensure files are stored in separate user folders, making them easier to find.

-----(1)

2. **Implement a Folder Structure within Each User Account:** Within each user account's folder, create a logical folder structure based on the person's activities, interests, or project types. For example:
    - **Documents:** For letters, reports, and other written work.
    - **Pictures:** For photos and images.
    - **Music:** For audio files.
    - **Projects:** For specific projects or work-related files.

-----(1)

(iii) *1 Mark* The reason it's important to indicate the source when reproducing content from a published book is to **give proper credit to the original author and avoid plagiarism**.
It acknowledges their intellectual property and respects their work.

-----(1)

(iv) *2 Marks* (a) One advantage of using SaaS is **lower upfront costs and reduced IT management overhead**.

- **Explanation:** With SaaS, you typically pay a subscription fee instead of buying and installing software. The provider handles maintenance, updates, and infrastructure, reducing the burden on your IT department.

-----(1)

(b) One disadvantage of using SaaS is **limited customization and control**.

- **Explanation:** SaaS applications are often designed for a broad audience, meaning you might not be able to customize them to perfectly fit your specific needs. You also have less control over the underlying infrastructure and data storage.

-----(1)

Part II, Q6 = 10 Marks

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7. (i) *4 Marks* (a) Answer already given as model answer.

(b) Select the piece of text → Click 6 → Choose font size 18

or

Select the piece of text → Click 11 until font size change into 18 -----(½)

(c) Select the sentence → Click 7 -----(½)

(d) Select the word → Click 9 -----(½)

(e) Select the number 2 → Click 12 -----(½)

(f) Click 4 → Choose three rows and two rows -----(½)

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(g) Click 15 → Type the contents one by one by pressing Enter key. -----(½)

(h) Select the piece of text → Click 14 -----(½)

(i) Place the cursor where you want to insert image → Click 1 →Choose Picture -----(½)

(ii) *1 Mark* Spell Check/Grammar Check -----(1)

(iii) *1 Mark* The thesaurus feature in a word processing software can be used to **find synonyms (words with similar meanings) or antonyms (words with opposite meanings)** for a selected word. This helps users:

- **Improve vocabulary:** By offering alternative words, it helps avoid repetition and makes writing more varied and interesting.
- **Find the right word:** It helps users choose the most appropriate word to convey their intended meaning.
- **Enhance clarity and style:** It can help improve the flow and readability of the text by offering more precise or eloquent word choices.

-----(1)

(iv) *2 Marks*
(a) Portrait -----(1)
(b) Landscape -----(1)

(v) *1 Mark* The benefit of the 'mail merge' facility in a word processing software is that it allows you to **create personalized documents (like letters, envelopes, labels, or emails) for multiple recipients at once.**

-----(1)

(vi) *1 Mark* The benefit of the 'Print Preview' feature in a word processing software is that it allows you to **see exactly how your document will look when printed, before you actually print it.**

-----(1)

Part II, Q7 = 10 Marks

Please send all your comments and feedbacks about my tentative marking scheme.

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